Factoring out the greatest common factor (GCF)

1 Find the greatest common factor between each set of terms:

a 3 and $15x$ **b** $8a$ and $9a$ **c** $33x$ and $6x^2$ **d** $12b$ and $15b^2x$

2 Complete the following statements:

a $4r + 12 = \square\,(r + 3)$ **b** $5v - 30 = \square\,(v - 6)$

c $18c - 24 = \square\,(3c - 4)$ **d** $2t^2 + 2t = 2t\,(\square + \square)$

3 Factor the following expressions:

a $6v + 30$ **b** $45t - 40$

c $-2s - 10$ **d** $-12s + 10$

e $y^2 + 4y$ **f** $k^2 - 5k$

g $2u^2 - 8u$ **h** $4t + 2t^2$

i $42x - x^2$ **j** $7x + 35$

k $36 - 6x$ **l** $-2x + 10$

m $-100 - 10x$ **n** $-18y^2 + 6$

o $9\,(3x + 4) + 4\,(3x + 4)$ **p** $8t\,(t - 3) + 9\,(t - 3)$

Additional questions

4 Xander was asked to factor the expression $35x^2y + 10xy^2 - 5xy$. Identify his error.
Xander:
The greatest common factor is $5xy$ so the factored form of the expression is $5xy(7x + 2y)$

5 A farmer wants to create a set of adjacent fields which all have the same width. He plans to create the smallest field in the shape of a square, with the largest field 9 times the size of the smallest field, and the middle field to have a length that is 5 units more than the smallest field.

a Write expressions for the area of each field.

b Use factoring to determine the dimensions of the entire field.

6 Explain how dividing monomials relates to factoring the greatest common factor.

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7 Explain why the greatest common factor of the variables in any expression has the *least* possible exponent of any of the terms.

8 You are to design a photo collage made of two large square photos with a side length of x and four smaller rectangular photos that have a height of x and a width of 4 inches.

a Find an algebraic expression for the area of the rectangle formed if the photos are all placed in a single row. Draw an example of what this arrangement would look like.

b Fully factor your answer from part (a) and then draw a photo arrangement that would match these dimensions.

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